

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 7 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 7

Subject Code & Name: BTCVE705B - FINITE ELEMENT METHOD

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question) 12

- 1 What is the primary advantage of the Finite Element Method (FEM) in engineering analysis?
 - a) It provides an exact solution to all problems.
 - b) It can handle complex geometries and boundary conditions.
 - c) It is faster than analytical solutions in all cases.
 - d) It does not require computational resources.
- 2 The Minimum Potential Energy Principle is commonly used in:
 - a) Fluid mechanics problems
 - b) Structural mechanics problems
 - c) Electrical circuit analysis only
 - d) Heat transfer problems
- 3 The Galerkin's Method is a special case of which method?
 - a) Collocation Method
 - b) Weighted Residual Method
 - c) Variational Method
 - d) Euler-Lagrange Method
- 4 The method of least squares minimizes:
 - a) The integral of the governing equation
 - b) The sum of squared residuals over the domain
 - c) The total potential energy
 - d) The number of elements in FEM
- 5 What is the primary unknown in a 1-D bar element subjected to axial loading?
 - a) Displacement
 - b) Stress
 - c) Strain
 - d) Load
- 6 What type of differential equation is typically solved using 1-D FEM?
 - a) Partial Differential Equation (PDE)
 - b) Algebraic Equation
 - c) Ordinary Differential Equation (ODE)
 - d) Integral Equation

- 7 The Galerkin's approach in finite element analysis is based on:
 - a) Minimum potential energy principle
 - b) Weighted residual method
 - c) Ritz method
 - d) Classical beam theory
- 8 In the formulation of a 1D beam element, which function is used to approximate the displacement?
 - a) Step function
 - b) Polynomial shape function
 - c) Trigonometric function
 - d) Exponential function
- 9 Which one of the following is correct for plane stress conditions?
 - a) Out-of-plane normal stress is zero
 - b) In-plane stresses are zero
 - c) Shear stress is neglected
 - d) Only axial loads are considered
- 10 The Constant Strain Triangle (CST) element assumes strain to be:
 - a) Linear throughout the element
 - b) Constant within the element
 - c) Quadratic within the element
 - d) Varying along the thickness
- 11 In 3D finite element analysis, which type of element is suitable for modeling curved surfaces?
 - a) Hexahedral element
 - b) Tetrahedral element
 - c) Beam element
 - d) Line element
- 12 In FEM, software like ANSYS, ABAQUS, and NASTRAN are used for:
 - a) Solving analytical equations
 - b) Finite element simulations
 - c) Pre-processing models
 - d) Post-processing results

Q2. Solve the following.

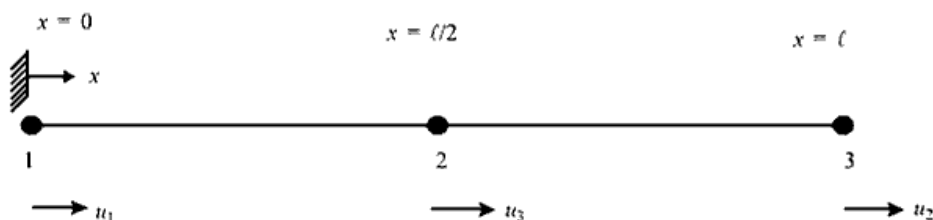
- A) Explain how the finite element method works. 6
- B) Calculate shape functions for the 1D quadratic element with three nodes. Use a local coordinate system: 6

Q3. Solve the following.

- A) Write the general procedure used to solve any problem by finite element analysis. 6
- B) Explain properties of shape functions. 6

Q4. Solve Any Two of the following.

- A) Write a note on Galerkin's approach of FEM. 6
- B) Explain the formulation of a 1-D beam element in FEM by the direct approach method. <https://www.batuonline.com> 6
- C) Determine the shape function for the three-noded quadratic bar element as shown in the figure: 6



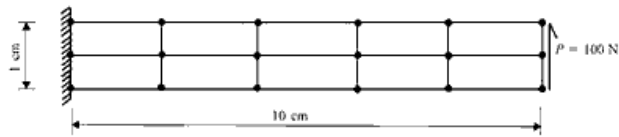
Q5. Solve Any Two of the following.

- A) Explain in detail CST Element (i.e., the three-noded triangular element). 6
- B) A bar is subjected to thermal stresses. Consider a uniform rod of length L , a cross-section A , Young's Modulus E , and the coefficient of thermal expansion α , heated uniformly through a temperature rise of T° C. Determine the thermal stresses in the bar for the bar having one end as a fixed end and the other as a free end. 6
- C) State and explain the finite element modelling of initial strains. 6

Q6. Solve Any Two of the following.

A) Explain 3-D stress analysis using FEM. 6

B) Consider the cantilever beam as shown in the below figure subjected to shear load at the tip. Use the 2D 6 plane elements to model the bending deformation. Under the tip calculate the expected tip deflection and slope. $EI = 0.1667 \times 10^{10} \text{ N-cm}^2$



C) Write the shape functions for the four-node rectangular element as shown in the figure. 6

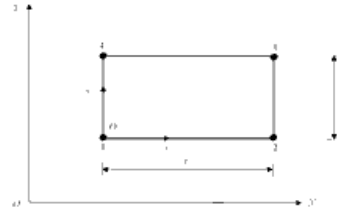


Fig. 5.28 A four-noded rectangular element.

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